UNIT42 · FIRMWAREX

Firmware Security Assessment

x86_64 controlled E2E firmware

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Executive summary

This assessment recorded **17 findings** across 2 audited binaries and 6 filesystem entries. Analysis status: **complete**.

Results describe the analysis performed on this image. They do not establish that the firmware is secure or that every candidate is exploitable. Severity and confidence are separate; no CVSS score is inferred.

Critical	High	Medium	Low	Info
0	1	6	10	0

Scope and coverage

Stage	Outcome	Restriction
extract	extracted	See scope limits below
static	completed	See scope limits below
emulation	completed	See scope limits below
symbolic	completed	See scope limits below
secrets crypto	completed	See scope limits below
normalize	completed	See scope limits below

Untrusted execution: blocked_pending_mentor_approval. Raw artifact access: disabled.

- Only registered controlled images execute; other images receive static and scanner analysis.
- Unknown path components are pseudonymized to prevent proprietary-string disclosure.
- Symbolic findings are bounded candidates for human review.

- Only executed stages contribute evidence. Blocked, skipped or incomplete analysis is not a clean bill of health.
- This is an automated firmware-image assessment, not a live-device or network scan.
- Recommendations are general review guidance, not verified fixes.

Binary protection summary

Binary	Architecture	NX	PIE	RELRO	Canary	FORTIFY
/bin/service-probe	x86_64	Present	Not detected	partial	Present	Not detected
/bin/taint-demo	x86_64	Present	Not detected	partial	Not detected	Not detected

Detailed findings

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1. Memory safety candidate High

Property	Result
CWE	CWE-787
Affected file	/bin/taint-demo
Architecture	x86_64
Confidence	probable
Review status	new
Address	0x500008
Function / sink	strcpy
Detected	2026-10-20T05:30:27.138658Z

Evidence

Observation	Value
condition satisfiable	true

Observation	Value
Observed taint path	0x401136 → 0x401040 → 0x500010 → 0x40115c → 0x40116b → 0x401030 → 0x500008
Reaching condition (diagnostic expression)	[{"op":"UGE","args":[{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract","args":[247,240, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":126,"bits":8}]},{"op":"UGE","args": [{"op":"Extract","args":[239,232, {"symbol":"fwx_cgi_stdin_0","bits":248}]}, {"constant":32,"bits":8}]},{"op":"ULE","args": [{"op":"Extract" … [excerpt; full expression in JSON export]
fault address	0x404030
overflow offset	8
stack hash	bfc322b46d82910dab6757cff82dd695801b2df2172325cb b73b4e88d0926fc2

Evidence reference: evidence:daadef28c2b437bba4117860e60976541b0b185026370254a4f86967405b62d7. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

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2. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established

Property	Result
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231273Z

Evidence

Observation	Value
rule	credential
fingerprint	9f6dfbbc6616a2b111899b503153248f0e40420ef3c496c501f4910e1d36edd9
byte offset	9
byte length	19

Evidence reference: evidence:da29f2ecaf6cb2720c1034ca91c7bcc1fdbc39947a97c85db6e01d30186f7541. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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3. Embedded secret candidate Medium

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231306Z

Evidence

Observation	Value
rule	credential
fingerprint	50cbe8769298797f275b9e0ccc1724c8761d2e440773b82773439b49d40e2a69
byte offset	37
byte length	24

Evidence reference: evidence:eb99bb87c7d3bd9de869e197e385530f42fd704fb4f99a81cca66cb3abb90b6a. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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4. Embedded secret candidate Medium

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231331Z

Evidence

Observation	Value
rule	private_key

Observation	Value
fingerprint	84aaf6a4be7a82452210d992023a9a353649e5b35f1b93ca3b2cf2c528723e0e
byte offset	62
byte length	78

Evidence reference: evidence:f4e26cfd6ec50f3b361c4d7069cbdb91892650555fd23adb0483e62ece3ee3d3. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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5. Weak cryptography candidate Medium

Property	Result
CWE	CWE-328
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231417Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	a1acd5c8533ed7767c8d8845aeef89b276e5bffd1d37c1fd812d2f4b530e8ba0

Observation	Value
byte offset	237
byte length	3

Evidence reference: evidence:4df88a859a34be05aa1db7c1a24c89070b8d067ec6ef33dee326ac0f3a7b0a3d. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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6. Weak cryptography candidate Medium

Property	Result
CWE	CWE-327
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231439Z

Evidence

Observation	Value
rule	security_weak_crypto
fingerprint	751521e7e526823307de978d01f205f690069ff4ad96ba72f3c9e8542bf39fc6
byte offset	255
byte length	3

Evidence reference: evidence:af56b39bbc0a212b3184e321ecd707727cc6ef961db4395eee470ae41012e3d4. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the cryptographic operation and its security context. Replace deprecated algorithms with an approved modern design after compatibility testing.

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7. Embedded secret candidate Medium

Property	Result
CWE	CWE-259
Affected file	/etc/redacted-713c77bbc24fe2b4
Architecture	x86_64
Confidence	probable
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231680Z

Evidence

Observation	Value
rule	credential
fingerprint	1710f8c0f2088c6dcf43120521be92d8e993003b4da1dbb043e488b349e57900
byte offset	38
byte length	25

Evidence reference: evidence:922a58b1c16b23ab2b32e07b3ada59e92819b05912b7bdf3de3fe655117ddb98. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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8. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-20T05:30:22.796180Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

9. Memory safety candidate Low

Property	Result
CWE	CWE-676
Affected file	/bin/taint-demo
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	strcpy
Detected	2026-10-20T05:30:22.796425Z

Evidence

Observation	Value
rule	potentially_dangerous_function_symbol
invocation proven	false
overflow proven	false

Evidence reference: evidence:9f26cf77847cb0f150b93bf6a32de0fe421d1fc84e000f539293d7be970a4c71. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Review the affected function and input bounds. Confirm the reported path before changing code; use length-checked operations and appropriate compiler protections.

10. Embedded secret candidate Low

Property	Result
CWE	CWE-798

Property	Result
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.230245Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	b8774025fef2064fb83b2d7f2e3d9b5e29e1fa1623f4e50241ffc7767a51c3f9
byte offset	511650
byte length	50

Evidence reference: evidence:79f2c922f408b11d2bdceb2e99010b3f621a198404b25e5d425ed139aa140e47. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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11. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64

Property	Result
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.230319Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	df143ca1dbf4b990a1c3a49617e003c33ec3a810fb489daac1d5048d16718f84
byte offset	591137
byte length	26

Evidence reference: evidence:90534c298875a8f0070a4bd56879ed285c91ac8f00ce46ebbbed354a90ff0cc3. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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12. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new

Property	Result
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.230349Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	c01c24d26d0d328977c736a9ad159a6bc22dd9c033d4238531678eb95bf65ae4
byte offset	591169
byte length	26

Evidence reference: evidence:6c9b9c3b25adc017a7fcee2fcf063c6330f7d3cd84b803b5ac0837d6ab05ed16. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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13. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable

Property	Result
Detected	2026-10-20T05:30:27.230379Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	cf43db5a89e60bcf06f7ff1908dd69e3a1a2b0a50d37b6b98316d722f2b3526e
byte offset	591328
byte length	36

Evidence reference: evidence:a7c7f6d0af51b4e208c23e8ccb990aa8a5e1f53a5938bf21edd8667f04c489a2. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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14. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/bin/service-probe
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.230405Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	87a1329eb49d0986f46549916e9bf51eba15e651a9dc2d32ddf9120c0ed6c059
byte offset	591392
byte length	36

Evidence reference: evidence:020821e29638de5171413e5298a81cc26866fda97f8c9bcbedf3404965689888. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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15. Embedded secret candidate Low

Property	Result
CWE	CWE-321
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231353Z

Evidence

Observation	Value
rule	certificate

Observation	Value
fingerprint	8ab906bd46dc80119cadd142e89258910966f146e41b289540285004d012a800
byte offset	141
byte length	75

Evidence reference: evidence:aacc5befe063a516277bd430c1c538691fc0ce52153a34e0776c67b71785ceeb. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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16. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231376Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	408c847260099765b5630514a6093a97a3321cfffeeae98e1d5472da36974fd4

Observation	Value
byte offset	29
byte length	32

Evidence reference: evidence:6fef4ca6da798aeb7ecffda79d6859c7780b96afd12854ac6c80744203899929. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

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17. Embedded secret candidate Low

Property	Result
CWE	CWE-798
Affected file	/etc/controlled.conf
Architecture	x86_64
Confidence	heuristic
Review status	new
Address	Not established
Function / sink	Not applicable
Detected	2026-10-20T05:30:27.231396Z

Evidence

Observation	Value
rule	high_entropy
fingerprint	4414f2ccabb7db5f2423ecb65bfbf9b0f1472033740f68f8770d758f0cfb0fe4
byte offset	259
byte length	29

Evidence reference: evidence:e4393b548662e6fa874f661244e424549e384aa914ee6123915531a9af6ac1f4. Secret values are withheld. A heuristic or probable finding is a candidate for human validation, not proof of exploitation.

Recommended next step

Verify whether the fingerprint represents an active credential or private key. If confirmed, remove it from the firmware and rotate the affected credential through your approved process.

Assessment identity

Property	Value
Firmware ID	af1031a8-c855-47a8-915a-07fa527d8e38
SHA-256	a71d1e4937d2eb468ea965462c27f6b3c99eb2207fa39bd59d04161bf45856c9
Job ID	87b6dfb2-775a-48c8-9d7d-a3e90147aa27
Image bytes	339968